

★ WEEK 15 · PART 3 · PRACTICE

Regimes and Breaks, Worked by Hand

Soft Regime Probabilities and Conviction Sizing

One Bayes step for a regime; SSE splits for a break

Part 3 at the trading desk, with the arithmetic shown. We turn a single return into a regime probability with one Bayes step, size the trade by conviction, then locate a structural break by comparing sums of squares.

3P

PART 3 · APPLICATIONS

Two Trading Decisions, by Hand

Case A: turn one -3% day into $P(\text{stress})$ with a Bayes step, then size by conviction instead of a hard label. Case B: find when a relationship broke by comparing split fits. Then a Claude Code lab.

What Part 3 Told Us

Now we make it concrete - with numbers

1

Soft regime probabilities beat hard labels

A Bayesian HMM reports the probability of each regime, not a flip-flopping 0/1 label. We compute one such probability by hand from a single day.

2

Breaks happen at uncertain dates

When the rules change once, the data tells us roughly when. We locate that date by comparing how well different splits fit.

3

Uncertainty flows to position size

Conviction sizing scales the trade by the regime probability. We compute soft sizes and contrast them with a hard on/off rule.

Everything here is a calculator exercise: one Bayes update, a few sums of squares, one sizing rule.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from a tiny, real-looking dataset.
- Compute the key number by hand, step by step.
- Then turn it into a trading decision.

What you should watch

- How one day updates a regime belief.
- How the fit locates a break in time.
- How the soft answer smooths the trade.

Soft Regimes and Conviction Sizing

One bad day should not flip you fully

A strategy works in calm markets and should stand down in stressed ones. A hard label snaps between full size and zero on a single threshold. Instead we compute the probability of stress from today's return, and let it scale the position smoothly.

The setup

- Two regimes: calm and stressed.
- The trade is favorable only in calm.
- Today brings one sharp return.

What we'll do by hand

- Update $P(\text{stress})$ with one Bayes step.
- Size the trade by conviction.
- Contrast it with a hard label.

The Regime Posterior

Bayes, applied to today's return

$$P(\text{stress} \mid r) = \frac{P(r|\text{stress}) P(\text{stress})}{P(r|\text{stress}) P(\text{stress}) + P(r|\text{calm}) P(\text{calm})}$$

Reading it

- Weigh each regime's prior by how well it explains today.
- A big move is far more likely under the stressed regime.
- Normalize, and you get $P(\text{stress})$ given the return.

Why it's soft

- The answer is a probability, not a label.
- It moves smoothly as evidence arrives.
- It carries its own confidence.

The Setup: Two Regimes

Priors, volatilities, and today's return

	Calm	Stressed
prior probability	0.80	0.20
daily volatility	1%	3%

Today's return is -3%. Goal: compute $P(\text{stress} \mid -3\%)$ with one Bayes step on the next slide.

P(stress) by Bayes

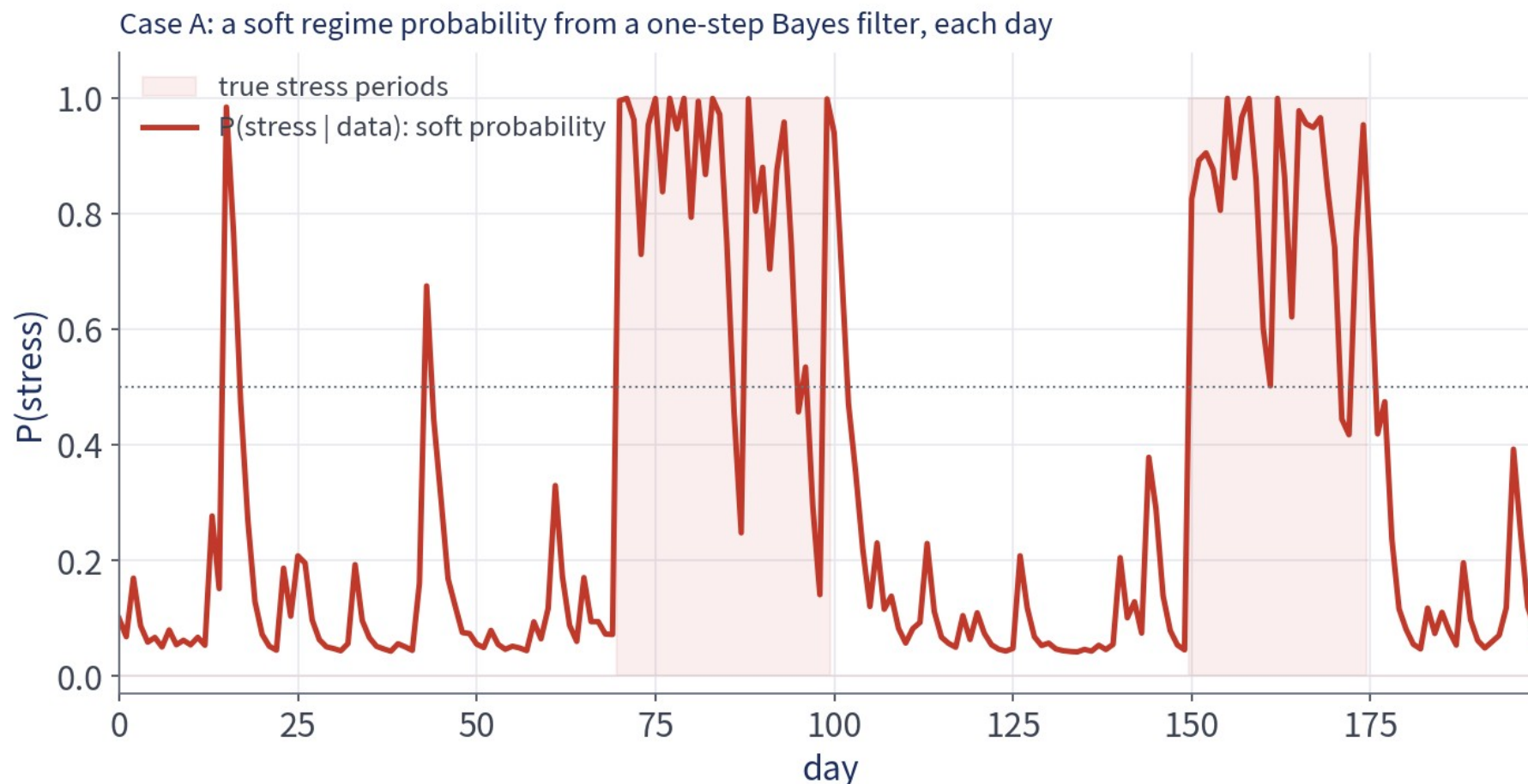
One update, three lines

$$\begin{aligned} \text{prior: } P(\text{calm}) &= 0.80, \quad P(\text{stress}) = 0.20; \quad \sigma_{\text{calm}} = 1\%, \quad \sigma_{\text{stress}} = 3\% \\ \text{today } r = -3\% : \quad L_{\text{calm}} &= \frac{1}{1}e^{-9/2} = 0.011, \quad L_{\text{stress}} = \frac{1}{3}e^{-9/18} = 0.202 \\ P(\text{stress} \mid r) &= \frac{0.202(0.20)}{0.202(0.20) + 0.011(0.80)} = \frac{0.0404}{0.0493} = \mathbf{0.82} \end{aligned}$$

A single -3% day lifts the stress probability from 0.20 to 0.82 - a strong but still soft signal, not a hard flip.

The Soft Regime Probability

Repeating that Bayes step every day gives a smooth $P(\text{stress})$ that rises into the stressed periods and falls out of them. It is a probability with shades of grey, never a jumpy 0/1 label.



Size by Conviction

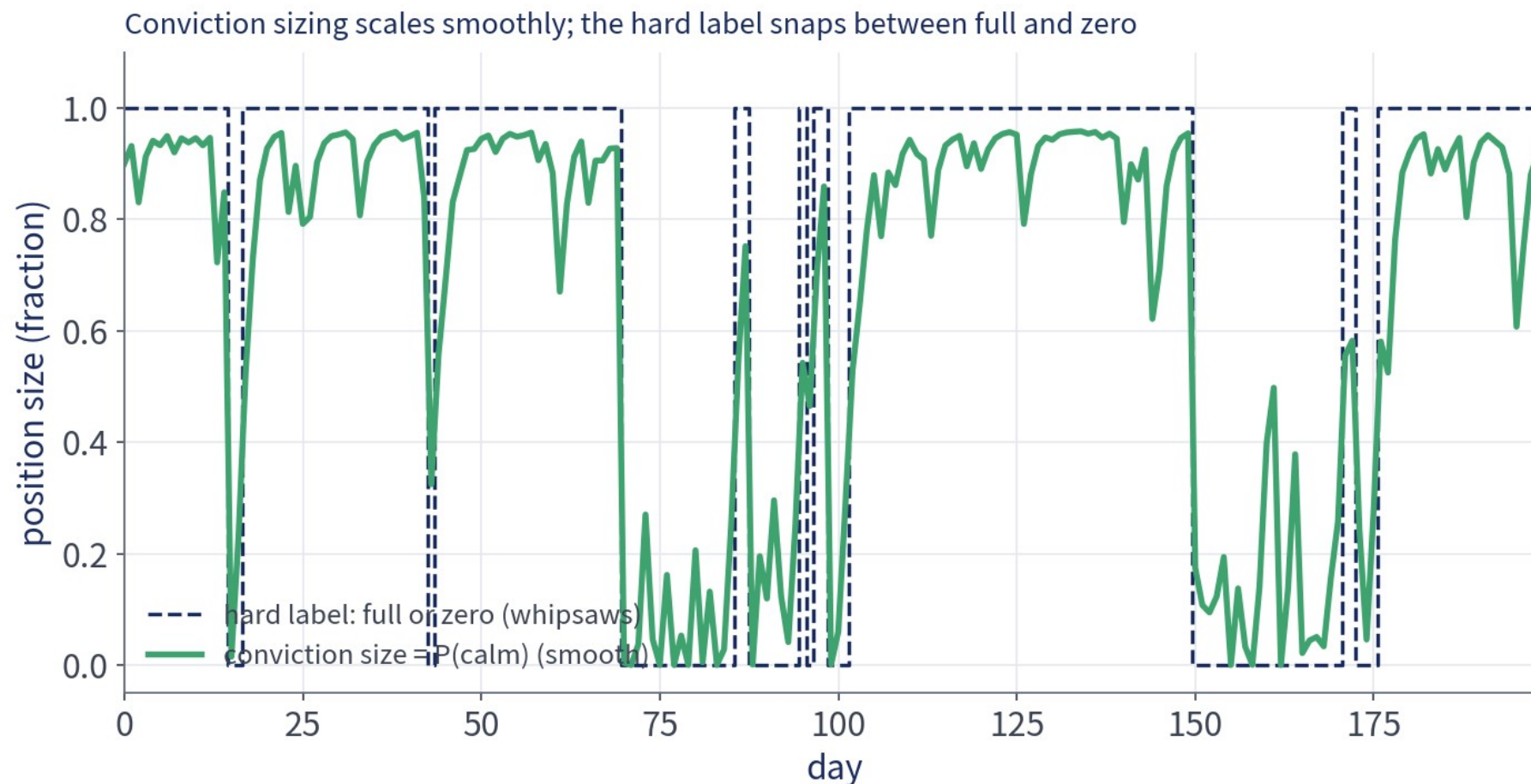
Soft probability vs hard label

$\text{size} = \text{edge} \times P(\text{favorable}), \quad \text{favorable} = \text{calm}$
before the -3% day: soft = 0.80, hard label = FULL
after ($P(\text{calm}) = 0.18$): soft = **0.18**, hard label = **ZERO**

The soft rule trims the position from 0.80 to 0.18 as stress rises. The hard label instead slams from full to zero on one day.

Conviction Sizing vs a Hard Label

The conviction size, equal to $P(\text{calm})$, glides up and down with the evidence. The hard label snaps between full and zero at the threshold - whipsaws in and out on borderline days and paying the costs each time.



What the Hard Label Costs

Whipsaw, turnover, and false certainty

With the hard label

- One threshold decides full or zero.
- Borderline days flip you in and out.
- Turnover and slippage pile up.

With conviction sizing

- The size tracks the probability smoothly.
- Borderline days get a middling weight.
- You trade less on genuine ambiguity.

When Did the Relationship Break?

A one-time shift at an unknown date

A trading relationship - a spread level, a strategy's mean - shifts once and stays shifted. We find WHEN by trying each possible break date and seeing which split fits best, then act on the post-break regime, sized by how sure we are of the date.

The setup

- A short series that jumps to a new level.
- The break date is unknown.
- Fitting one mean to all of it fits badly.

What we'll do by hand

- Compare no-break vs break fits.
- Score each candidate break date.
- Turn the scores into a posterior.

The Data: Six Months

Find where the level shifts

Month	1	2	3	4	5	6
Value (%)	+0.5	-0.2	+0.9	+1.7	+2.4	+2.0

Goal: does one mean fit, or did the level jump? We compare sums of squared errors (SSE) for each split.

No Break vs a Break

Compare how well each fits

no break: one mean 1.22, $SSE = 4.87$
break after month 3: $m_1 = 0.40$, $m_2 = 2.03$, $SSE = 0.87$
the break model fits about $5.6 \times$ better $\Rightarrow$ a break is real

Splitting after month 3 cuts the SSE from 4.87 to 0.87 - a far better fit. A break is real; now, exactly when?

Which Date? A Posterior

Score every candidate break

SSE if break after month 2, 3, 4 : 1.45, **0.87**, 1.97

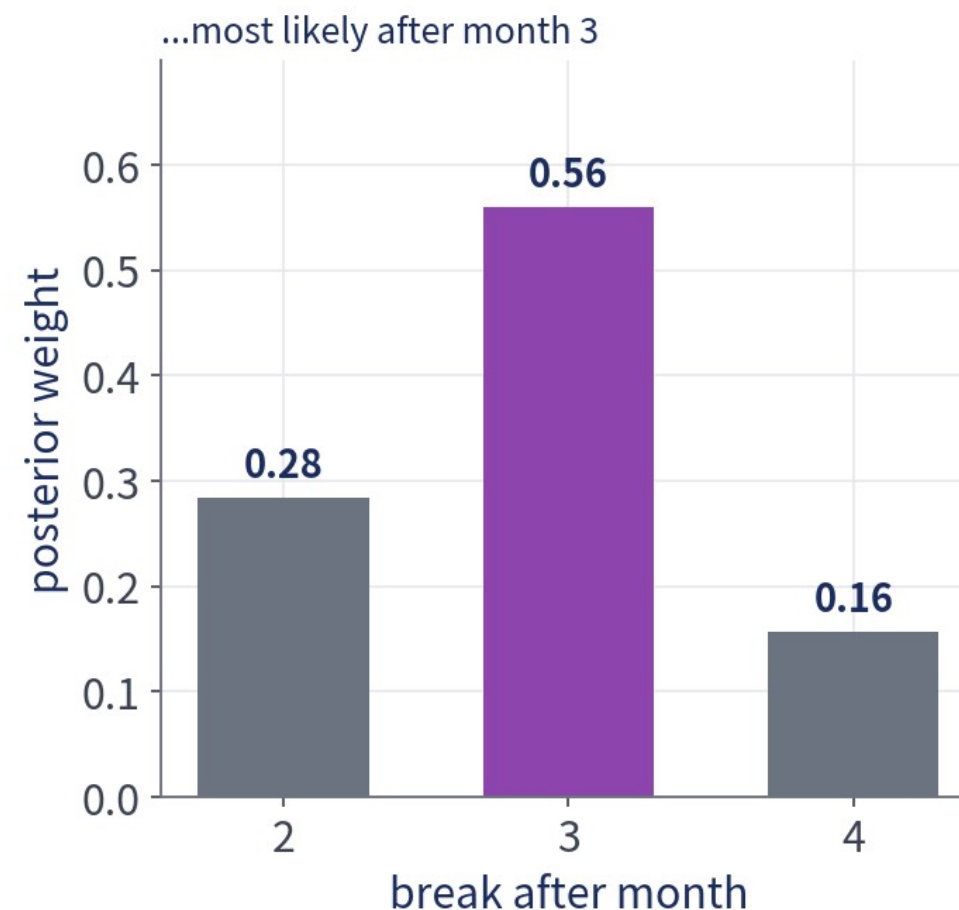
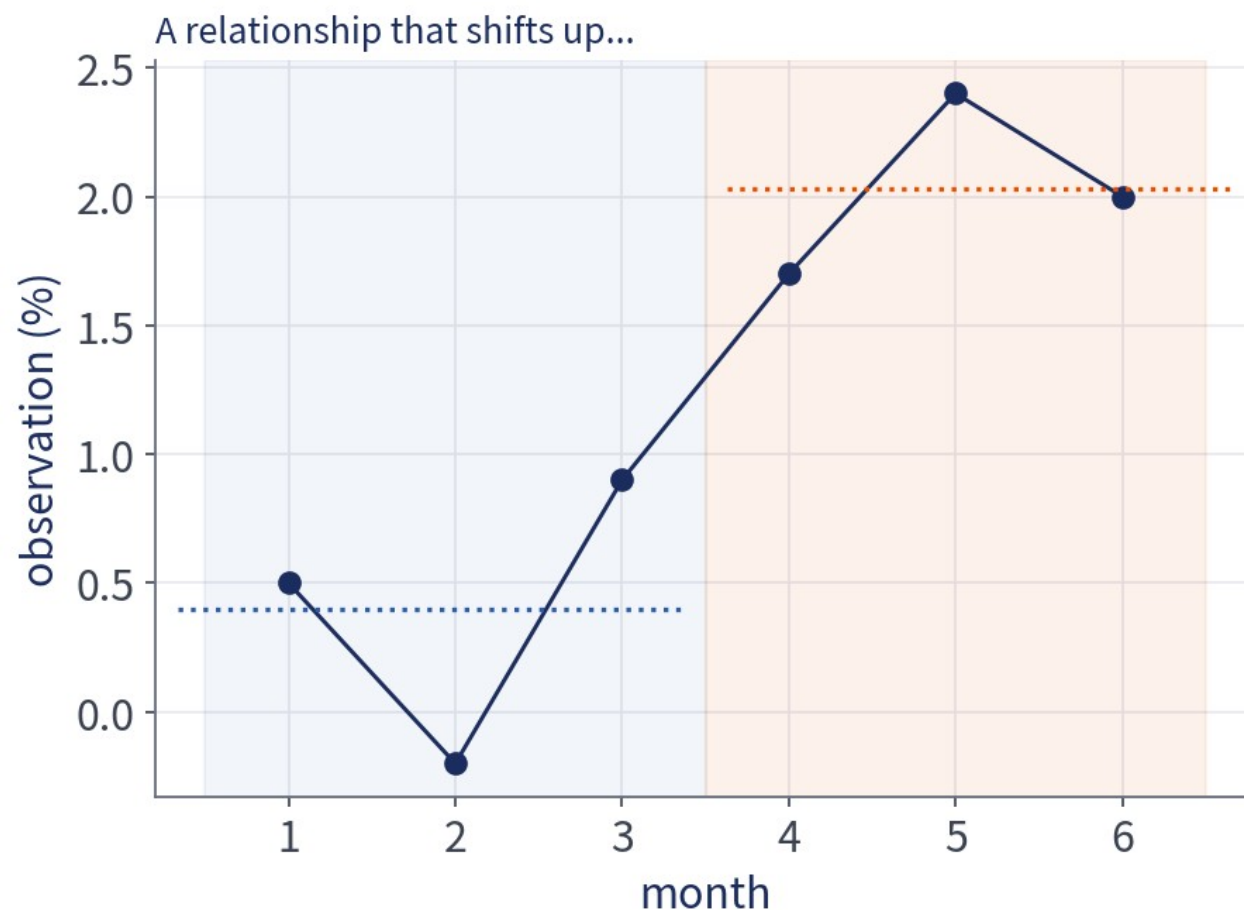
turn into weights $\propto e^{-\text{SSE}/2s^2}$: 0.28, **0.56**, 0.16

most likely after month 3 — but month 2 is plausible too

The best split is after month 3 (weight 0.56), but month 2 keeps a real 0.28. The break date itself is uncertain - and honestly so.

A Break With an Uncertain Date

The series shifts up around month 3 (left). Scoring each split turns into a posterior over the break date (right): it peaks after month 3 but spreads to month 2, telling us roughly - not exactly - when the rules changed



Act on the Break

Re-estimate, and size by certainty

Re-estimate after the break

- Use only the post-break data for the new regime.
- Old data would drag the estimate toward the past.
- The mean is now 2.03, not the blended 1.22.

Size by certainty of the date

- A sharp break date means act with confidence.
- A fuzzy one means blend across possible dates.
- Uncertainty about WHEN flows into the trade.

What Ignoring the Break Costs

Trading a stale relationship

If you ignore it

- You keep using the old, blended average.
- Your signal is centered on a level that's gone.
- Every trade is biased toward the past regime.

If you detect it

- You switch to the post-break parameters.
- Your signal reflects the current regime.
- You wait for more data when the date is unclear.

The Label-Switching Trap

Before you average regime estimates

What can go wrong

- When you fit regimes, their names are arbitrary.
- The sampler can swap 'calm' and 'stressed' mid-run.
- Averaging the raw draws then blends the two.

The simple fix

- Impose an order, e.g. calm has the lower volatility.
- That pins the labels down.
- Always check before trusting a regime average.

Conviction Sizing, in One Line

Both cases feed this

$$\text{position}_t \propto \underbrace{\text{edge}_t}_{\text{signal}} \times \underbrace{P(\text{favorable} \mid y)}_{\text{conviction}} \times \underbrace{\frac{1}{\text{band width}}}_{\text{less doubt, more size}}$$

The three inputs

- The edge: the raw strength of the signal.
- The conviction: $P(\text{favorable regime})$, from Case A.
- The certainty: tighter when the break date is clear.

The behavior

- Full size only when all three agree.
- Automatic caution when any is shaky.
- Uncertainty sets the bet, not a footnote.

The Common Pattern

Two calculations, one shape

Both cases, computed by hand, turn a soft belief into a sized action. A Bayes step gives a regime probability; a fit comparison gives a break date and its uncertainty. In each, that uncertainty flows straight into how much you trade.

The trap

- A hard yes/no on a regime or a break.
- Full commitment on a borderline call.
- Whipsaw and bias when the call is wrong.

The fix

- Compute a probability, not a label.
- Locate a break with its own uncertainty.
- Size the trade to that honest confidence.

Reproduce the Hand Calcs, Then Scale Up

Direct Claude Code to check your arithmetic

GOAL Verify both worked examples in code, then extend them to full series.

[INPUT] Prompt · Claude Code

Write and run `part3_practice.py` that:

CASE A - soft regime:

- * from priors 0.8/0.2 and vols 1%/3%, do one Bayes step on $r = -3\%$; print $P(\text{stress})$;
- * run the filter over a series and size by $P(\text{calm})$;
- * compare to a hard-label rule (turnover, whipsaw).

CASE B - break:

- * take the 6-month series; compute SSE for no break and for a break after months 2, 3, 4;
- * turn the SSEs into break-date weights;
- * re-estimate the post-break mean.

Match every printed number to the slides.

Check against the slides

- $P(\text{stress} \mid -3\%) = 0.82$.
- Soft size 0.80 $\rightarrow$ 0.18; hard full $\rightarrow$ zero.
- Break SSE 4.87 $\rightarrow$ 0.87 (after month 3).
- Date weights 0.28 / 0.56 / 0.16.

Deliverable: `part3_practice.py` whose printed numbers match the hand calculations, then the charts.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 How does one -3% day raise $P(\text{stress})$ to 0.82?

**Bayes: weigh priors
by each regime's fit**

Q2 How do you locate the break date?

**The split with the
lowest SSE**

Q3 Why size by $P(\text{favorable})$ instead of a hard label?

**It scales smoothly
and avoids whipsaw**

All three come from the Bayes step, the SSE comparison, and conviction sizing.

Three Pitfalls to Avoid

Where these applications slip

Common mistakes

- "A hard regime label is decisive, so it's better."
- "The break date is known once I see the jump."
- "I can average raw regime draws directly."

The reality

- A soft probability avoids whipsaw and states its doubt.
- The date has a posterior - month 2 was plausible too.
- Fix label switching first, or the regimes blur.

Two Calculations to Remember

Part 3 applications in one screen

1

One Bayes step turns a return into a regime probability

Priors 0.80/0.20 and vols 1%/3% take a -3% day to $P(\text{stress}) = 0.82$. Sizing by $P(\text{calm})$ trims the trade from 0.80 to 0.18 smoothly, where a hard label would slam from full to zero.

2

Comparing fits locates a break - and its uncertainty

Splitting after month 3 cut the SSE from 4.87 to 0.87. The break-date weights (0.28, 0.56, 0.16) say 'most likely month 3, but month 2 is plausible' - honest uncertainty about when.

3

Soft beliefs become sized actions

Conviction sizing multiplies edge by the regime probability and by certainty about the break. Uncertainty is not a footnote; it sets the bet - the heart of trading with a posterior.

Next - Week 16: Synthesis and the Final Project, where the whole course comes together.